

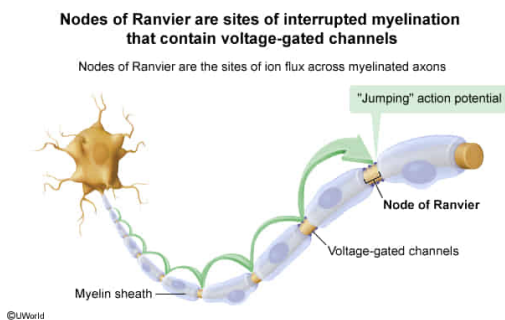
Which statement best describes the function of the nodes of Ranvier? The nodes of Ranvier:

- ☐ A. prevent depolarization from spreading toward the axon soma. [6%]
- ☐ B. promote entry of extracellular K^+ into unmyelinated axons. [6%]
- ☐ C. prevent entry of extracellular Na^+ into myelinated axons. [8%]
- ✓ ☒ D. increase the rate of depolarization toward the axon terminal. [78%]

Correct

78% answered correctly

Explanation:



An **action potential** is generated in a **neuron** when its cell body (soma) depolarizes enough that the membrane potential reaches a **threshold** for the opening of **voltage-gated channels**. At this triggering potential, the voltage-gated channels open in a coordinated fashion, allowing a series of ionic movements across the plasma membrane to occur. These movements allow the rapid maximal **depolarization** of a small area of the membrane due to Na^+ transport into the cell, followed by rapid **repolarization** as the Na^+ channels inactivate (close) and K^+ channels open, allowing K^+ to leave the cell.

Action potentials "move" **along an axon** toward the axon terminal as successive regions are depolarized, and the manner in which action potentials move is influenced by myelin. Myelin is formed by the plasma membrane from **glial cells**. Specifically, portions of glial cell plasma membrane are wrapped many times around an axon, thereby insulating the axon and preventing ion flow across the axon membrane. The wave of ion transport is **continuous** along unmyelinated axons but "jumps" in a discontinuous fashion along myelinated axons due to the insulation provided by myelin.

The rate at which a neural impulse is conveyed to the axon terminal is influenced by axon size and myelination: Larger-diameter axons and axons that are myelinated transmit neural impulses faster than smaller or unmyelinated axons, respectively. This advantage is partly due to myelination preventing the loss of current along the axon.

The question asks for the role of nodes of Ranvier in the transmission of neural impulses. **Nodes of Ranvier** are **small interruptions in the myelin sheath**. By exposing a short segment of unmyelinated axon, the nodes of Ranvier **promote entry of extracellular Na^+ into axons** while limiting the total number of ion channels that must open, thereby greatly increasing the rate at which neural impulses are conducted along myelinated axons.

(Choice A) Inactivation of voltage-gated Na^+ channels (*not* nodes of Ranvier) prevent depolarization from spreading back toward the axon soma.

(Choice B) Nodes of Ranvier are only present in *myelinated* axons.

(Choice C) Entry of extracellular Na^+ is promoted (*not* prevented) by nodes of Ranvier.

Educational objective:

Nodes of Ranvier are interruptions in the myelination of myelinated axons. By providing only periodic sites of Na^+ entry and loss of insulation, nodes of Ranvier allow much faster neural transmission in myelinated axons than in unmyelinated axons of similar size.

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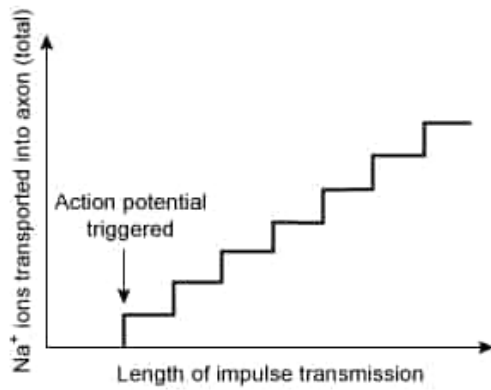
Subject:
Biology

QID:404169

System:
Endocrine and Nervous Systems

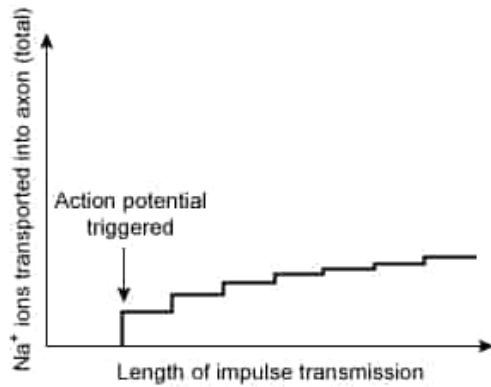
Which of the following graphs illustrates the likely changes in the number of Na^+ ions transported into a myelinated axon during transmission of a neural impulse along its length?

✓ ☒ A.



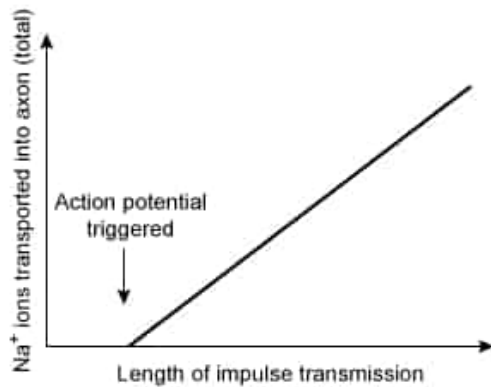
[38%]

☐ B.



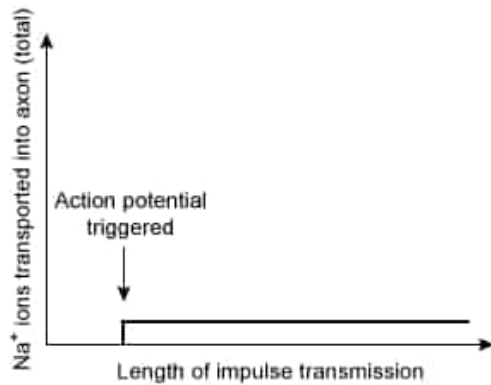
[29%]

☐ C.



[16%]

☐ D.

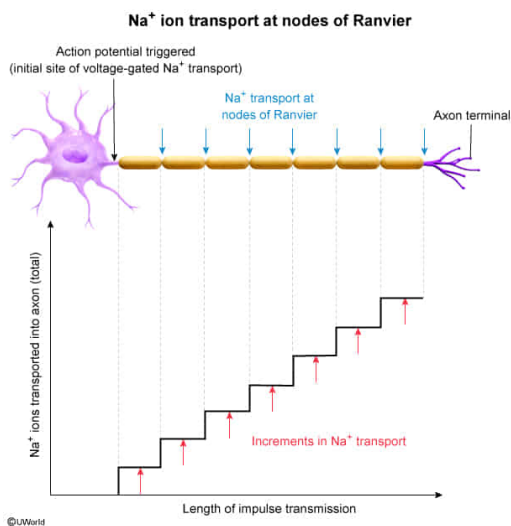


[15%]

Correct

37% answered correctly

Explanation:



Myelinated neurons have a layer of **myelin** insulation wrapped around their axons. This layer is produced by **glial cells** and is nearly continuous along the length of an axon, only interrupted at small, regularly spaced sites called **nodes of Ranvier**.

The presence of nodes of Ranvier greatly increases the rate of neural impulse transmission along myelinated nerve fibers. This occurs in part due to the reduced number of **action potentials** and associated membrane channel openings that must occur along the axon length. Furthermore, a diminishing of current occurs along the axon due to the insulation provided by myelin.

The question requires identifying the graph that illustrates the likely changes in the number of Na⁺ ions transported into a myelinated axon as a neural impulse is transmitted along the length of the axon. In myelinated fibers, **voltage-gated Na⁺ channels** are only present at nodes of Ranvier. As a result, the number of Na⁺ ions transported into a myelinated fiber **increases in a "stair-step" pattern**, with **increments** of approximately **equal size** in the **number of transported Na⁺ ions** at each node of Ranvier. Only the graph shown in Choice A depicts this pattern.

(Choice B) In a myelinated nerve fiber, the number of Na⁺ ions transported into the fiber during transmission of a neural impulse along its length increases intermittently, with increments occurring at each node of Ranvier. However, the magnitude of depolarization is maintained (*not* diminished) as the impulse is transmitted along the axon.

(Choice C) This graph depicts the relationship between Na⁺ transport and length of impulse transmission in an *unmyelinated* axon, in which the number of transported Na⁺ ions rises continuously along the length of the fiber as an impulse is transmitted.

(Choice D) Na⁺ ions are transported into myelinated axons as neural impulses are transmitted along their length; however, this graph incorrectly shows that Na⁺ transport only occurs at the site where the initial action potential was triggered.

Educational objective:

Nodes of Ranvier are interruptions in the myelination of myelinated axons. By providing only periodic sites of Na⁺ entry and loss of insulation, nodes of Ranvier allow markedly faster neural transmission in myelinated axons than in unmyelinated axons of similar size.

Time spent: 0 sec
Subject:
Biology

QID: 404170
System:
Endocrine and Nervous Systems